# BRYAN TALAVERA

**Data Scientist & Analyst** | UCSD B.S. Data Science, 2024

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## SUMMARY

UCSD B.S. Data Science graduate with hands-on experience in machine learning, dbt/Snowflake/Airflow pipelines, and Tableau dashboards. Built end-to-end projects from scratch, including a tuned XGBoost classifier on 408k+ NYC taxi trips (83% accuracy) and a Random Forest churn classifier (0.95 AUC). Targeting full-time data analyst or data engineer roles.

## EDUCATION

**University of California, San Diego · B.S. Data Science, Minor in Cognitive Science** June 2024

## CERTIFICATIONS

**Google Advanced Data Analytics Professional Certificate** · Coursera · 2024 · View credential

## PROJECTS

**NYC Taxi ELT Pipeline** | *Python · Airflow · dbt · Snowflake · AWS S3 · Docker · GitHub Actions* *In progress*
*github.com/Bryan-Tal/nyc-taxi-elt*

- Designed a batch ELT pipeline ingesting 100M+ NYC TLC trip records into Snowflake using Apache Airflow 2.10 DAGs and a Medallion architecture (Raw to Staging to Marts) modeled in dbt
- Secured cross-cloud authentication via AWS IAM role assumption and least-privilege Snowflake RBAC, separating functional and access roles
- Containerized the orchestration layer with Docker and configured GitHub Actions CI/CD for repeatable, idempotent deployments

**Predicting Employee Churn** | *Python · Scikit-learn · XGBoost · Pandas · GridSearchCV* *2025*
*github.com/Bryan-Tal/Predicting_Employee_Churn*

- Compared Decision Tree, Random Forest, and XGBoost classifiers on an 11,991-row HR dataset; selected Random Forest as champion via 5-fold GridSearchCV refit on ROC-AUC
- Achieved 0.95 AUC, 97.97% accuracy, and 90% recall on the held-out test set; engineered an overworked feature (>175 hrs/mo) that ranked among the top predictors
- Surfaced that 100% of employees with 7 concurrent projects left the company, identifying a workload threshold for HR retention strategies

**Predicting NYC Taxi Gratuities** | *Python · XGBoost · Scikit-learn · Pandas · GridSearchCV* *2025*
*github.com/Bryan-Tal/Predicting_NYC_Taxi_Gratuities*

- Built a binary classifier predicting generous tippers (>20%) on 408,000+ NYC yellow taxi trips, tuning XGBoost via 5-fold GridSearchCV refit on F1
- Achieved 83.3% accuracy and 84% F1 on the held-out test set, beating the Random Forest baseline by +12.7 F1 points
- Engineered time-of-day, day-of-week, and rate-code features from raw timestamps; documented cash-payment survivorship bias as a model limitation

**Predicting House Prices with OLS & Ridge Regression** | *Python · NumPy · SciPy · Pandas · Matplotlib* *2025*
*github.com/Bryan-Tal/Predicting_House_Prices_OLS*

- Implemented Multiple Linear Regression and Ridge Regression from scratch using the closed-form normal equation in NumPy, foregoing scikit-learn for the regression itself
- Validated all 5 OLS assumptions (Shapiro-Wilk, Anderson-Darling, K-S, VIF) and searched 4 scaling methods × 7 Ridge α values; champion model achieved $R^2 = 0.66$, RMSE = 0.108

**Med-Dash · UC San Diego Capstone** | *React · MongoDB · ECharts · Vital API* *Sep 2023 to Mar 2024*

- Built a personalized, interactive dashboard for patients to visualize daily health metrics (heart rate, sleep, activity) ingested from the Vital API
- Collaborated with a 4-person interdisciplinary team to integrate data persistence (MongoDB) and frontend visualization (React + ECharts); led validation across multiple wearable sensor types

## SKILLS

**Languages:** Python, SQL, JavaScript, R
**Machine Learning & Statistics:** Scikit-learn, XGBoost, Pandas, NumPy, SciPy, Statistical Testing (Shapiro-Wilk, Anderson-Darling, K-S, VIF), GridSearchCV
**Data Engineering:** Apache Airflow, dbt, Snowflake, AWS S3 & IAM, Docker, GitHub Actions
**Visualization & BI:** Tableau, Matplotlib, Seaborn, ECharts, D3.js
**Tools & Databases:** Git/GitHub, Jupyter, MongoDB, PostgreSQL, MS SQL Server

## WORK EXPERIENCE

**In-Store Shopper, Pavilions · San Diego, CA** Present

- Fulfilled 30+ time-sensitive grocery orders per shift while maintaining a 95%+ on-time completion rate
- Trained 4 new associates on workflow procedures and order-fulfillment best practices, reducing handoff delays during shift transitions
- Coordinated across 3 store departments to resolve stock discrepancies and customer escalations